

Scalability & Future Plan

Date	06 July 2026
Project Title	Rising Waters
Maximum Marks	1 Marks

Scalability & Future Plan :

Current system Limitations :

S.No	Limitation	Impact	Priority to Address
1	Uses historical dataset only	May not reflect real-time weather conditions	High
2	Runs on local Flask server	Limited accessibility for users	Medium
3	No user authentication or alert system	Limited security and notification features	Medium

Scalability plan :

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited local users	Deploy on cloud (Render/AWS) with multi-user support
2	Data Storage	Static Excel dataset	Integrate MySQL/PostgreSQL database with live data
3	Performance	Single-machine execution	Optimize model and enable server-side caching
4	Security	Basic Flask application	Add user authentication, HTTPS, and role-based access

Future Road map :

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy application on cloud	1 Month	Accessible from anywhere
Phase 2	Integrate real-time weather and rainfall APIs	2 Months	More accurate flood prediction
Phase 3	Add SMS/Email alerts and GPS-based location detection	3 Months	Faster warning and improved user safety
Phase 4	Develop Android mobile application with interactive dashboard	4–6 Months	Better usability and wider adoption